A COFILTERED LIMIT OF NONEMPTY SPECTRAL SPACES WITH CLOSED TRANSITION IMAGES IS NONEMPTY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that the limit of nonempty quasi-compact Hausdorff spaces over a cofiltered category is nonempty. Invoked inside the counted proof (source lines 5045-5047). Provides nonemptiness after passing to the constructible topologies.
- The statement that the constructible topology of a spectral space is quasi-compact Hausdorff, and that spectral maps are continuous for it. Used inside the counted proof (source lines 5040-5044). Puts the inverse system in the setting of the previous nonemptiness result.

Source reference. Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-constructible-hausdorff-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 4522–4522. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-hausdorff-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-nonempty-limit: sources/stacks-topology/pinned-original-source.tex, source lines 2724–2724. Navigation pointer only: the counted text refers to the Stacks item lemma-nonempty-limit, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the

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Lemma 0.1. *Let $\mathcal{I}$ be a cofiltered category. Let $i \mapsto X_i$ be a diagram of spectral spaces such that for $a : j \to i$ in $\mathcal{I}$ the corresponding map $f_a : X_j \to X_i$ is spectral.*

(1) *Given nonempty subsets $Z_i \subset X_i$ closed in the constructible topology with $f_a(Z_j) \subset Z_i$ for all $a : j \to i$ in $\mathcal{I}$, then $\lim Z_i$ is nonempty.*
(2) *If each X_i is nonempty, then $X = \lim X_i$ is nonempty.*

Proof. Denote X_i' the space X_i endowed with the constructible topology and Z_i' the corresponding subspace of X_i'. Let $a : j \to i$ in $\mathcal{I}$ be a morphism. As f_a is spectral it defines a continuous map $f_a : X_j' \to X_i'$. Thus $f_a|_{Z_j} : Z_j' \to Z_i'$ is a continuous map of quasi-compact Hausdorff spaces (by Lemmas lemma-constructible-hausdorff-quasi-compact and lemma-closed-in-quasi-compact). By Lemma lemma-nonempty-limit the space $\lim Z_i'$ is nonempty. Since $\lim Z_i' = \lim Z_i$ as sets we conclude. □